# Generating pairs from characters and subgroup incidence

Human Compression project

**Abstract**

We give a common exact calculation for generating pairs whose product is an involution. One subgroup-incidence matrix serves all prescribed element orders. A complementary stabilizer construction produces orbit representatives and tests simultaneous inversion. For the O'Nan group we verify two class obstructions used by Connor and Leemans. The numerical census of generating pairs is not inferred from the character table alone and remains outside this component.

## 1 The objects and the retained character identity

For a finite group $H$ and integers $p, q > 1$, write

$$X_H(p,q) = \{(R,S) \in H^2 : |R| = p,\ |S| = q,\ |RS| = 2\},$$
$$E_H(p,q) = \{(R,S) \in X_H(p,q) : \langle R, S\rangle = H\}.$$

Put $x_H(p,q) = |X_H(p,q)|$ and $e_H(p,q) = |E_H(p,q)|$. Pairs are ordered. Restricting a table to $p \le q$ chooses one order for the types; it does not further identify a pair with its dual when $p = q$. Any such additional quotient requires a separate orbit calculation.

We retain the ordinary character theorem of Frobenius, in the form quoted by Connor–Leemans [1], Theorem 2.1. If $C_i$ are the conjugacy classes of $H$, with representatives $c_i$, then the number of triples in $C_i \times C_j \times C_k$ with product one is

$$\frac{|C_i||C_j||C_k|}{|H|} \sum_{\chi \in \mathrm{Irr}(H)} \frac{\chi(c_i)\chi(c_j)\chi(c_k)}{\chi(1)}.$$

For each character and order define

$$v_{\chi,s} = \sum_{|c_i| = s} |C_i|\chi(c_i).$$

Summing the retained identity over the three order classes gives

$$x_H(p,q) = \frac{1}{|H|} \sum_{\chi} \frac{v_{\chi,2} v_{\chi,p} v_{\chi,q}}{\chi(1)}. \tag{1}$$

Indeed $(R,S)$ corresponds bijectively to $((RS)^{-1}, R, S)$ with product one, and $(RS)^{-1} = RS$ here. All quantities are computed exactly in the character field; the resulting nonnegative integer is checked, not rounded. The vectors $v_{\chi,s}$ can be formed once for all desired $(p,q)$.

## 2 One incidence calculation for every type

Choose representatives $H_1, \ldots, H_m$ of every conjugacy class of subgroups of a fixed finite group $G$, ordered by nondecreasing size. For each $i, j$ let

$$M_{ij} = |\{K \le H_i : K \text{ is conjugate in } G \text{ to } H_j\}|.$$

This is a precise weighted subgroup input. A list of subgroup names or unweighted containment relations alone does not supply $M$.

**Theorem 1.** *The matrix $M$ is unit lower triangular. For every pair of orders,*

$$x_{H_i}(p,q) = \sum_j M_{ij} e_{H_j}(p,q), \qquad e_{H_i}(p,q) = x_{H_i}(p,q) - \sum_{j<i} M_{ij} e_{H_j}(p,q).$$

*Consequently the same integral triangular calculation recovers the whole vector of generating-pair counts from the corresponding all-pair vectors.*

*Proof.* Every pair in $X_{H_i}(p,q)$ generates a unique subgroup $K \leq H_i$. Its contribution belongs to exactly one of the sets $E_K(p,q)$. Conjugation preserves both orders and the product relation, and gives a bijection $E_K(p,q) \simeq E_{H_j}(p,q)$ whenever $K$ is conjugate to $H_j$. Partitioning by the actual generated subgroup proves the first formula. If $K \leq H_i$ and $|K| \geq |H_i|$, then equality of orders forces $K = H_i$. There is precisely one such subgroup in its own conjugacy class and none in a different equal-order class. Thus $M$ is unit lower triangular, and moving the preceding terms gives the second formula. The same proof applies in every coordinate of a vector of types; $M$ is independent of those types. □

**Lemma 2.** *The action of* $\mathrm{Aut}(H)$ *on* $E_H(p,q)$ *is free. Its number of orbits is* $e_H(p,q)/|\mathrm{Aut}(H)|$.

*Proof.* An automorphism fixing both generators fixes every word in them and hence every element of $H$. Every stabilizer is therefore trivial, so every orbit has the indicated group order. This freeness need not hold on $X_H(p,q)$; dividing the all-pair count by $|\mathrm{Aut}(H)|$ is not the required orbit count. □

Theorem 1 is Hall's subgroup inversion arranged as one common calculation. It is not a new inversion theorem. Formula (1) and Lemma 2 likewise retain the methods attributed in [1]. The source already reuses subgroup data in its algorithm; the vector organization supplies an explicit shared interface, not a claim that its authors recomputed every subgroup for each arithmetic operation.

## 3 Constructing the representatives and testing inversion

Let $A \leq \mathrm{Aut}(G)$ be a specified automorphism group, and let $J$ be the set of nonidentity involutions in $G$. The map

$$(R,S) \longmapsto (S,RS), \qquad (s,t) \longmapsto (ts^{-1}, s)$$

is an $A$-equivariant bijection between product-involution pairs and $G \times J$. Moreover $\langle R,S\rangle = \langle s,t\rangle$.

**Theorem 3.** *Choose one representative $s$ from every $A$-orbit of elements of order $q$. For each such $s$, choose representatives for the point stabilizer $A_s$ on*

$$J_{s,p} = \{t \in J : |ts^{-1}| = p\}.$$

*Retain exactly those with $\langle s,t\rangle = G$. The resulting pairs $(ts^{-1}, s)$ represent each $A$-orbit in $E_G(p,q)$ exactly once.*

*Proof.* Every pair has its second coordinate moved to a chosen $s$ by an element of $A$. Its new involution coordinate belongs to $J_{s,p}$. This set is $A_s$-invariant, because its definition uses only $s$, multiplication and element order. A further element of $A_s$ moves that coordinate to a chosen representative. Generation is also invariant, proving coverage after the final test. If two retained pairs are equivalent, their second coordinates first lie in the same chosen orbit and are therefore equal. The transporting automorphism then lies in $A_s$, so their chosen involution coordinates are equal. This proves uniqueness. □

This is the same stabilizer argument used in the acquired L071 enumeration proof, applied to a different equivariant set. It does not require IP of a tuple or that $s$ be an involution. Using only $\langle s\rangle$-orbits on $J$, as in the source's constructive route, still covers the candidates but requires its subsequent full-isomorphism removal. The full stabilizer avoids that particular redundancy. Neither construction may use an unproved expected count as its stopping certificate.

For a generating pair with $p, q > 2$, at most one $\theta \in \mathrm{Aut}(G)$ can simultaneously send $(R, S)$ to $(R^{-1}, S^{-1})$. Indeed the ratio of two such automorphisms fixes the generating pair. If it exists then $\theta^2 = 1$ and $\theta \neq 1$. The simultaneous inversion criterion for regular maps is retained from [1], Section 5: no such automorphism gives the chiral case, inner inversion gives the non-orientable case for the centreless simple group, and outer inversion gives the extension case. Testing a pair's two elements separately for invertibility is only a necessary condition, since the same automorphism must do both.

## 4 The two O'Nan obstructions

Let $G$ be the O'Nan simple group, and acquire its named ordinary character table and that of $\mathrm{Aut}(G) = G : 2$ from the Atlas data in CTblLib. We also acquire the stored fusion from $G$-classes to $G : 2$-classes. These finite-group identifications and tables are retained inputs, not identified solely from the displayed orders.

**Lemma 4.** *Let $G \trianglelefteq A$ have index two. Let $i$ be a conjugacy class of $G$, write $\iota(i)$ for its inverse class and $f(i)$ for its class in $A$. An element of class $i$ has an inner inverter precisely when $\iota(i) = i$. It has an inverter in $A \setminus G$ precisely when either*

$$\iota(i) \neq i, \quad f(\iota(i)) = f(i), \quad \textit{or} \quad \iota(i) = i, \quad |f^{-1}(f(i))| = 1.$$

*Proof.* The first assertion is the definition of conjugacy to the inverse in $G$. Every $A$-class contained in $G$ is the union of either one or two $G$-classes, since $G$ has index two. If the inverse class is different but fuses with $i$, an inverter exists in $A$ and none exists in $G$, so it is outside $G$. If the inverse class is $i$ and the fusion fibre has two classes, every outer conjugation exchanges them and cannot invert the element. If the fibre has one class, an outer conjugate can be brought back to the element by inner conjugation, giving an outer centralizer element. Composing it with an inner inverter gives an outer inverter. These cases exhaust the possibilities. □

The exact table calculation gives the following data; the last columns count classes whose elements admit the specified inverter.

| order | $G$-classes | inner inverter | outer inverter |
|---|---|---|---|
| 8 | 2 | 2 | 0 |
| 16 | 4 | 4 | 0 |
| 20 | 2 | 0 | 2 |
| 31 | 2 | 0 | 2 |

Hence no generating pair with an element of order 20 or 31 admits inner inversion; no pair with an element of order 8 or 16 admits outer inversion. This recovers exactly Lemmas 5.2 and 5.3 of [1], using the retained simultaneous-inversion interface. It does not prove any positive type or the remaining isolated exclusions in Theorem 1.1.

## 5 Verification and the remaining input

The exact programs compare character counts with all literal pairs in $S_3, A_4, S_4, A_5$. A separate control forms the complete conjugate-subgroup incidence matrices for $S_3, D_8, A_4, S_4, A_5$,

performs the vector inversion, and compares every resulting count against direct subgroup generation. These finite controls test the implementation; the uniform results are the preceding proofs, not an extrapolation from the examples.

For O'Nan, the program computes all character counts and the complete inverse/fusion data. They are only all-pair counts. The full weighted subgroup incidence and subgroup character inputs required for Theorem 1, or a completed representative enumeration under Theorem 3, have not been supplied here. In particular this component does not certify the source's 796772 generating-pair orbits, its 136 numerical table entries or its full chirality spectrum. No whole-original admission, new conjecture, Lean statement or compression measurement is assigned to it.

## References


[1] T. Connor and D. Leemans, Algorithmic enumeration of regular maps, *Ars Math. Contemp.*10 (2016),211–222.

[2] P. Hall, The Eulerian functions of a group, *Quart. J. Math. Oxford*7 (1936),134–151.

[3] G. Frobenius, Ueber Gruppencharaktere, *Sitzungsber. Preuss. Akad. Wiss. Berlin* (1896),985–1021.

[4] J.H. Conway, R.T. Curtis, S.P. Norton, R.A. Parker and R.A. Wilson, *Atlas of Finite Groups*, Oxford University Press,1985.